

D.A.V. LOHAGHAT
PRE-BOARD EXAMINATION II-2023-24

SUBJECT - PHYSICS

MM: 70

CLASS-XII

Time: 3:00 hours

General Instructions :

Read the following instructions very carefully and strictly follow them :

1. This question paper contains **33** questions. **All** questions are **compulsory**.
2. This question paper is divided into **five** Sections **A, B, C, D** and **E**.
3. In **Section A** Questions no. **1** to **16** are Multiple Choice (MCQ) type questions, carrying **1** mark each.
4. In **Section B** Questions no. **17** to **21** are Very Short Answer (VSA) type questions, carrying **2** marks each.
5. In **Section C** Questions no. **22** to **28** are Short Answer (SA) type questions, carrying **3** marks each.
6. In **Section D** Questions no. **29** to **31** are Long Answer (LA) type questions carrying **5** marks each.
7. In **Section E** Questions no. **32** and **33** are case-based questions carrying **4** marks each.
8. There is no overall choice. However, an internal choice has been provided in 3 questions in Section B, 4 questions in Section C, and 3 questions in Section D . You have to attempt only one of the choices in such questions.
9. Use of calculators is **not** allowed.
10. Use the following values of physical constants, if required :

$$c = 3 \times 10^8 \text{ m/s}$$

$$h = 6.63 \times 10^{-34} \text{ Js}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-2}$$

$$\text{Mass of electron } (m_e) = 9.1 \times 10^{-31} \text{ kg}$$

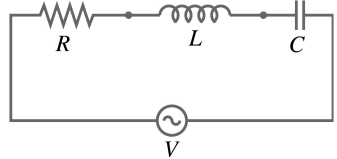
$$\text{Mass of neutron} = 1.675 \times 10^{-27} \text{ kg}$$

$$\text{Mass of proton} = 1.673 \times 10^{-27} \text{ kg}$$

$$\text{Avogadro's number} = 6.023 \times 10^{23} \text{ per gram mole}$$

$$\text{Boltzmann constant} = 1.38 \times 10^{-23} \text{ JK}^{-1}$$

Section A - MCQ

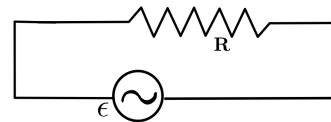
- Q 1.** If the velocity of electron in the ground state of hydrogen atom is v , it's velocity in the second excited state would be: [1]
- A. $2v$ B. $\frac{v}{2}$ C. $3v$ D. $\frac{v}{3}$
- Q 2.** A storage battery of emf 8.0 V and internal resistance $0.5\ \Omega$ is being charged by a 120 V DC supply using a series resistor of $15.5\ \Omega$ What is the terminal voltage of the battery during charging? [1]
- A. 10 V B. 11 V C. 11.5 V D. 8 V
- Q 3.** A solenoid of length 0.5 m has a radius of 1 cm and is made up of 500 turns. It carries a current of 5 A. What is the magnitude of the magnetic field inside the solenoid? [1]
- A. $\pi \times 10^{-3}$ T B. $2\pi \times 10^{-3}$ T C. $3\pi \times 10^{-3}$ T D. $4\pi \times 10^{-3}$ T
- Q 4.** Current sensitivity of a galvanometer can be increased by decreasing: [1]
- A. Magnetic field B C. Torsional constant K
B. Number of turns N D. Area A
- Q 5.** Figure shows a series LCR circuit connected to a variable frequency 230 V source. $L = 5.0$ H, $C = 80\ \mu F$, $R = 40\ \Omega$ Determine the source frequency which drives the circuit in resonance? [1]
- 
- A. 150 rad/s B. 75 rad/s C. 50 rad/s D. all frequencies.
- Q 6.** The relative permeability of a substance X is slightly less than unity and that of substance Y is slightly more than unity, then [1]
- A. X - paramagnetic and Y- ferromagnetic C. X and Y both are paramagnetic
B. X- diamagnetic and Y- ferromagnetic D. X- diamagnetic and Y- paramagnetic
- Q 7.** A tank is filled with a liquid to a height of 12.5 cm. The apparent depth of a needle lying at the bottom of the tank is measured by a microscope to be 9.4 cm. What is the refractive index of the liquid? [1]
- A. 1.55 B. 0.75 C. 1.23 D. 1.33
- Q 8.** The energy E of a hydrogen atom is given by $E = -\frac{13.6}{n^2}$ eV. The energy ejected when the electron jumps from $n = 3$ state to $n = 2$ state of hydrogen is approximately- [1]
- A. 0.85 eV B. 1.5 eV C. 1.9 eV D. 3.4 eV
- Q 9.** In Young's double-slit experiment using monochromatic light of wavelength λ , the intensity of light at a point on the screen where path difference is λ is K units. What is the intensity of light at a point where path difference is $\frac{\lambda}{3}$? [1]

A. K

B. $\frac{K}{2}$ C. $\frac{K}{4}$

D. Zero

- Q 10.** A pure resistor is connected to an AC power source as shown below. Which of the following statement(s) is/are TRUE?



[1]

- I:** The average current flowing through the circuit during one full cycle is zero.
II: The current in the resistor leads the voltage by $\frac{\pi}{2}$.
III: The average power dissipated by the resistor is zero.

A. only I

B. only I and II

C. only II and III

D. All- I, II and II

- Q 11.** At what rate does the electric field change between the plates of a square capacitor of side 5 cm, if the plates are spaced 1.2 mm apart and the voltage across them is changing at a rate of 60 V/s?

[1]

A. $7.2 \times 10^{-2} \text{ Vm}^{-1}\text{s}^{-1}$ C. $12 \times 10^2 \text{ Vm}^{-1}\text{s}^{-1}$ B. $30 \times 10^{-1} \text{ Vm}^{-1}\text{s}^{-1}$ D. $5 \times 10^4 \text{ Vm}^{-1}\text{s}^{-1}$

- Q 12.** The objective of a telescope must be of large diameter in order to

[1]

- A. remove chromatic aberration
 B. remove spherical aberration and high magnification
 C. gather more light and for high resolution
 D. increase its range of observation

Questions number **13** to **16** are Assertion (A) and Reason (R) type questions. Two statements are given - one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (a), (b), (c) and (d) as given below.

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
 (b) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).
 (c) Assertion (A) is true, but Reason (R) is false.
 (d) Assertion (A) is false and Reason (R) is also false.

- Q 13. Assertion (A) :** The electric field at every point is normal to the equipotential surface passing through that point.

[1]

Reason (R) : No work is required to move a test charge on an equipotential surface.

- Q 14. Assertion (A) :** Kinetic energy of photo electrons emitted by a photosensitive surface depends upon the intensity of incident photon.

[1]

Reason (R) : The ejection of electrons from metallic surface is possible with frequency of incident photon below the threshold frequency.

Q 15. Assertion (A) : Among the particles of same kinetic energy, lighter particles have greater de-Broglie wavelength. [1]

Reason (R) : The de-Broglie wavelength of a particle depends only on the charge of the particle.

Q 16. Assertion (A) : If a dielectric is inserted between the plates of a charged capacitor which is disconnected from the battery, then charge on the capacitor remains the same. [1]

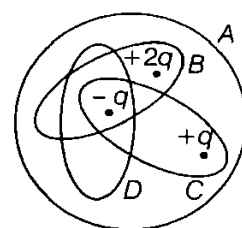
Reason (R) : Charge on isolated system remains conserved.

Section B - Very Short Answer Questions

Q 17. (a) Two capacitors of capacitance of $6 \mu F$ and $12 \mu F$ are connected in series with a battery. The voltage across the $6 \mu F$ capacitor is 2 V. Compute the total battery voltage. [2]

OR

(b) Rank the Gaussian surfaces as shown in the figure. In order of increasing electric flux, starting with the most negative. [2]



Q 18. In a plane electromagnetic wave, the electric field oscillates sinusoidally at a frequency of 2.0×10^{10} Hz and amplitude 48 Vm^{-1} [2]

- i. What is the wavelength of the wave?
- ii. What is the amplitude of the oscillating magnetic field?

Q 19. (a) A closely wound solenoid of 800 turns and area of cross section $2.5 \times 10^{-4} \text{ m}^2$ carries a current of 3.0 A. Explain the sense in which the solenoid acts like a bar magnet. What is its associated magnetic moment? [2]

OR

(b) A short bar magnet has a magnetic moment of 0.48 JT^{-1} . Give the direction and magnitude of the magnetic field produced by the magnet at a distance of 10 cm from the centre of the magnet on (a) the axis, (b) the equatorial lines (normal bisector) of the magnet. [2]

Q 20. (a) Find the radius of curvature of the convex surface of a plano-convex lens, whose focal length is 0.3 m and the refractive index of the material of the lens is 1.5 [2]

OR

(b) A telescope consists of two lenses of focal lengths 20 cm and 5 cm. Obtain its magnifying power when the final image is (i) at infinity (ii) at 25 cm from the lenses of eye. [2]

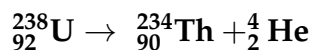
Q 21. If light of wavelength 412.5 nm is incident on each of the metals given below, which ones will show photoelectric emission and why?

Metal	Work Function (eV)
Na	1.92
K	2.15
Ca	3.20
Mo	4.17

Section C - Short Answer Questions

Q 22. Calculate the energy released in MeV in the following nuclear reaction:

[3]



Use the information below to answer the question:

$$\text{Mass of } {}_{92}^{238}\text{U} = 238.05079 \text{ amu}, \quad \text{Mass of } {}_{90}^{234}\text{Th} = 234.043630 \text{ amu}$$

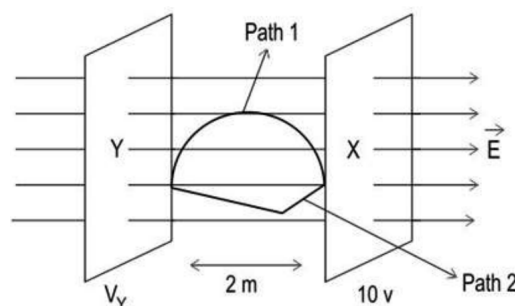
$$\text{Mass of } {}_2^4\text{He} = 4.002600 \text{ amu} \quad 1 \text{ u} = 931.5 \text{ MeV}/c^2$$

Q 23. X and Y are two equipotential surfaces separated by a distance of **2 m** in a uniform electric field of **10 V/m**. Surface X has a potential of **10 V**

[3]

(a) Calculate the potential of surface Y.

(b) What is the work done in moving a **+2 C** charge from surface **Y** to surface **X** along **path 1**? How will this work change when the charge is moved along **Path 2**? Give a reason for your answer.



Q 24. (a) What is the shape of the wave front in each of the following cases:

[3]

- i. Light diverging from a point source.
- ii. Light emerging out of a convex lens when a point source is placed at its focus.
- iii. The portion of the wave front of light from a distant star intercepted by the Earth.

OR

(b) Derive an expression for the fringe width of interference pattern obtained in Young's double slit experiment.

Q 25. (a) What is electric flux? Write its S.I unit. Using Gauss's theorem deduce an expression for the electric field at a point (both inside and outside) due to a uniformly charged spherical shell.

[3]

OR

(b) Derive an expression for the electric field E due to a dipole of length $2a$ at a point, distant r , from the centre of the dipole, on the axial line. Draw a graph of E v/s r for $r \gg a$.

[3]

Q 26. (a) If $f = 0.5 \text{ m}$ for a glass lens, what is the power of the lens?

[3]

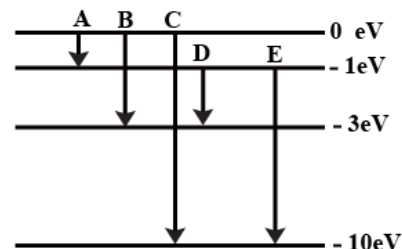
(b) A convex lens has 20 cm focal length in air. What is focal length in water? (Refractive index of air-water = 1.33, refractive index for air-glass = 1.5.)

Q 27. (a) Draw a schematic arrangement of Geiger- Marsden experiment. Calculate the distance of closest approach when a 7.7 MeV α -particle approaches a gold nucleus ($Z = 79$)

[3]

OR

- (b) i. The energy levels of an atom of element X are shown in the diagram. Which one of the level transitions will result in the emission of photons of wavelength 620 nm? support your answer with mathematical calculations.



[2½]

- ii. name the hydrogen spectral series that can be observed in the visible region.

[½]

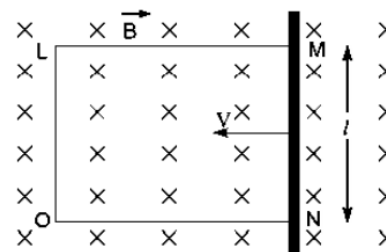
Q 28. (a) State Faraday's laws of electromagnetic induction.

[3]

- (b) A metallic rod of 1 m length is rotated with a frequency of 50 rev/s, with one end hinged at the centre and the other end at the circumference of a circular metallic ring of radius 1 m, about an axis passing through the centre and perpendicular to the plane of the ring. A constant uniform magnetic field of 1 T parallel to the axis is present everywhere. What is the emf between the centre and the metallic ring?

OR

- (c) A rectangular conductor LMNO is placed in a uniform magnetic field of 0.5 T. The field is directed perpendicular to the plane of the conductor. When the arm MN of length 20 cm is moved towards left with a velocity of 10 ms^{-1} , calculate the emf induced in the arm. Given the resistance of the arm to be 5 ohm (assuming that other arms are of negligible resistance), find the value of the current in the arm.



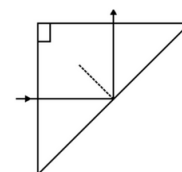
[3]

Section D - Long Answer Questions

- Q 29.** (a) i. Plot a graph to show variation of the angle of deviation as a function of angle of incidence for light passing through a prism. Derive an expression for refractive index of the prism in terms of angle of minimum deviation and angle of prism.

[3½]

- ii. A ray of light incident normally on one face of a right isosceles prism is totally reflected as shown in fig. What must be the minimum value of refractive index of glass? Give relevant calculations.



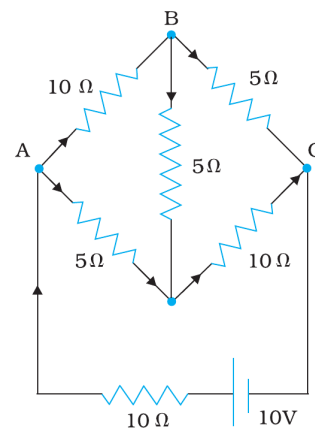
[1½]

OR

- (b) i. A Young's double slit setup is illuminated with monochromatic light. If the intensity of light passing through one of the slits is reduced, explain the changes that can be seen in the appearance of the bright and dark fringes?
- ii. α) A single slit diffraction setup is illuminated with green light of wavelength **500 nm**. If the width of the slit is **1 mm** and the screen is **2 m** away from the slits, calculate the width of the central maximum.
- β) What will happen to the width of the central maximum, if the green light is replaced with the red light? Give a reason for your answer.
- iii. What is the effect on the interference fringes in Young's double slit experiment, if the separation between the two slits is decreased? Justify your answer.

[5]

- Q 30.** (a) i. State the two Kirchhoff's laws.
 ii. Determine the current in each branch of the network shown in Fig.

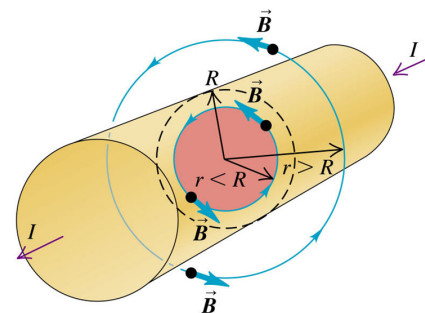


OR

- (b) i. Define the term drift velocity.
 ii. On the basis of electron drift, derive an expression for resistivity of a conductor in terms of number density of free electrons and relaxation time. On what factors does resistivity of a conductor depend?
 iii. Why alloys like constantan and Manganin are used for making standard resistors?

- Q 31.** (a) i. State and explain Ampere's circular law?

- ii. Figure shows a long straight wire of a circular cross-section (radius R) carrying steady current I . The current I is uniformly distributed across this cross-section. Calculate the magnetic field in the region $r < R$ and $r > R$. Also plot the magnitude of $\vec{B}$ with distance r from the centre of the wire.



OR

- (b) i. With the help of a neat and labelled diagram, explain the underlying principle and working of a moving coil galvanometer.
 ii. what is the function (a) uniform radial field (b) Soft iron core in such a device.

Section E - Case Study

- Q 32.** The instantaneous voltage, $V = V_0 \sin \omega t$

Then the instantaneous value of current $I(t)$ in the circuit at instant 't' is given by $I(t) = \frac{V_0}{1} \sin \left(\omega t + \frac{\pi}{2} \right)$. The capacitive

reactance limits the amplitude of current in a purely capacitive circuit and it is given by $X_C = \frac{1}{\omega C}$

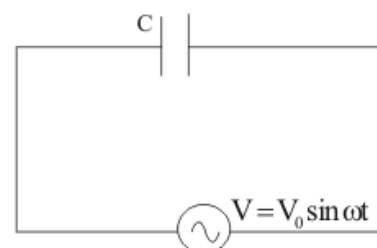


Figure 7. A.C. Circuit containing only capacitor

- (a) Find the capacitive reactance of a $5 \mu\text{F}$ capacitor for a frequency of 10^6 Hz .
 (b) Draw a graph showing the variation of capacitive reactance X_C with frequency.
 (c) $1 \mu\text{F}$ is joined to a 200 V , 50 Hz alternator. Find the rms current through capacitor.
 (d) What is the value of average power over a complete cycle in this case?

Q 33. Read the following paragraph and answer the questions that follow.

The number of charge carriers can be changed by 'doping' of a suitable impurity in pure semiconductors. Such semiconductors are known as **extrinsic semiconductors**. These are of two types (n-type and p-type). In **n-type semiconductors**, $n_e \gg n_h$ while in **p-type semiconductors** $n_h \gg n_e$. **n-type** semiconducting Si or Ge is obtained by doping with pentavalent atoms (donors) like As, Sb, P, etc., while **p-type** Si or Ge can be obtained by doping with trivalent atom (acceptors) like B, Al, In etc. $n_e n_h = n_i^2$ in all cases. Further, the material possesses an overall charge neutrality.

When an external voltage (V) is applied across a semiconductor diode such that p-side is connected to the positive terminal of the battery and n-side to the negative terminal it is said to be **forward biased**. When the diode is forward biased the depletion layer width decreases and the barrier height is reduced. The effective barrier height under forward bias is $(V_0 - V)$.

When an external voltage(V) is applied across the diode such that n-side is positive and p-side is negative, it is said to be **reverse biased**. During reverse bias the barrier height increases and the depletion region widens due to the change in the electric field. The effective barrier height under reverse bias is $(V_0 + V)$

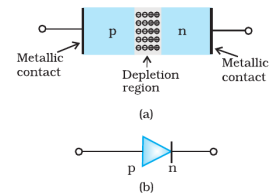


FIGURE 14.12 (a) Semiconductor diode, (b) Symbol for p-n junction diode.

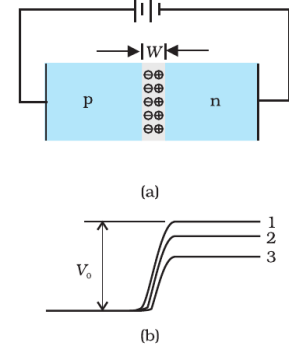


FIGURE 14.13 (a) p-n junction diode under forward bias, (b) Barrier potential (1) without battery, (2) Low battery voltage, and (3) High voltage battery.

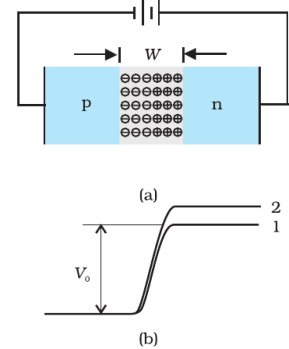
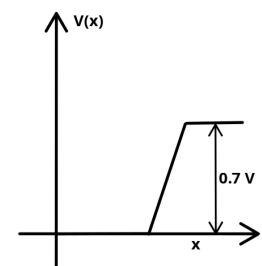


FIGURE 14.15 (a) Diode under reverse bias, (b) Barrier potential under reverse bias.

(a) A semiconductor has equal electron and hole concentration of $2 \times 10^8 \text{ m}^{-3}$. On doping with a certain impurity, the hole concentration increases to $4 \times 10^{10} \text{ m}^{-3}$.

- What type of semiconductor is obtained on doping?
- Calculate the new electron concentration of the semiconductor.

(b) The graph of the potential barrier (V) vs width of the depletion region (x), when a diode is unbiased at room temperature, is shown below. Plot a comparative graph of the potential barrier (V) vs width of the depletion region (x) of the diode when it is reversed biased. Give reason.



(c) When a forward bias is applied to a p-n junction, it

- raises the potential barrier.
- reduces the majority carrier current to zero.
- lowers the potential barrier.
- None of the above.